

Bio-treatment of Anthroquinone Textile Dyes (Reactive Blue-19, Brilliant Blue-R, Acid Blue-25) using Soil Fungi Isolated from Dye Effluents.

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Abstract: Decolorization of textile dyes (Remazol Brilliant Blue-R, Brilliant Blue-R and Acid Blue-25) by soil fungi *Aspergillus terreus*, *Aspergillus flavus* strain AJ, *Aspergillus fumigatus* strain YHYS, *Aspergillus tamarii* strain SRRC 1088, *Aspergillus niger* strain MCAS2, *Pencillium* sps isolated from the dye amended samples collected around the textile industries and tested for their efficiency in decolourization of Anthraquinone textile dyes. It was observed that all the six fungal species are found to be efficient in decolourization. Among the six fungal species *Aspergillus fumigatus* strain YHYS showed maximum decolorization (86.3%) with AB-25 and *Aspergillus terreus* show very low decolorization activity (46.2%) with BBR. The optimization of decolourization was tested with carbon source (89.9%) with AB-25, nitrogen source (86.3%) with AB-25, P^H 5.5 observed high decolorization activity (86.3%) with AB-25 at 37^oc temperature, dye concentration and agitation. High efficiency of dye decolorization is a key for the degradation of polluted effluents. Therefore these isolates were treated as promising species for Anthroquinone dye decolorization.

Key words: Textile dyes, decolourization, optimization, *Aspergillus* and *Pencillium* sps.

1. Introduction

Anthroquinone dyes are aromatic compounds the most important largest class of synthetic dyes used in commercial applications. Synthetic dyes are being used by several textile industries intensively because of their ease and cost effectiveness. During the dyeing process, approximately 10-15% of the dyes used are released into the wastewater the presences of these dyes in the aqueous ecosystem are the cause of serious environmental and health concern. In general the textile dyes are highly reactive and therefore during process it is very difficult to treat. So that the commonly used dyes by the textile industry are Synthetic, Azo and Anthraquinone etc (Reactive Blue-19, Brilliant Blue-R and Acid Blue-25).Most of the textile industries utilize large volume of water during wet processing operation and there by generates substantial quantities of waste water is the principle route by which dyestuffs enters into the soil environment (Elliott, 1999).In recent years, the treatment of waste to develop a new biological processes including aerobic and anaerobic microorganisms used for degradation and also

reutilization process (Francis et al. 2009). During the past decade, the use of microbiological methods for degradation and decolorization have been under active development in textile and dyestuff industry (Knapp et al., 1995). Microbial decolorisation and degradation is an environmental friendly and cost competitive to chemical decomposition (Verma & Madamvar, 2003). Bioremediation depends on the pollution removal process with microorganisms has much attention as microbial decolorisation (Kalyani et al., 2008). Among the microorganism's bacteria are the most commonly used for various bioremediation processes. But fungi are concerned most of the reports on bioremediation of textile dyes from the environment by fungi are scanty. There are few reports available fungal decolorization studies especially white rot fungi are the most commonly used for bioremediation. Anaerobic degradation of dyes also leads to the formation of highly carcinogenic and mutagenic. This study represents the isolation of aerobic fungi from soil and identified the fungal strain that are shown to be highly effective for decolorization of Anthroquinone dyes and evaluate their performance under various conditions of carbon, nitrogen, temperature, pH etc. There are many microorganisms especially fungi which are abundant in the environment. The fungi has the capacity to degrade wide variety of hazardous chemicals in and around has arisen the interest to use them for bioremediation (Brar et al., 2006). It can be seen from the literature *Phanerochaete chrysosporium*, basidiomycetes is the only choice of organism reported to be potential decolourizer of the effluent treatment (Krik et al., 1992). In the present study an attempt has been made to utilize the common soil inhabiting fungi isolated indigenously from the soil contaminated by industrial effluents for decolourization of Anthroquinone textile dyes.

2. Materials and Methods

The soil samples collected near the textile industry where the effluents are discharged from the dye industries (Chirala, Bapatla (A.P) and Siera Silk Mills Ltd., Bangalore). The collected samples were analysed for fungi by serial dilution and inoculated onto Potato dextrose agar medium plates and placed them for incubation at $30\pm 1^\circ\text{C}$ for 2 days. Isolated fungal species were identified by using microscope.

The isolated fungal species sequenced by 18s RNA and identified as *Aspergillus terreus*, *Aspergillus flavus* strain AJ, *Aspergillus fumigatus* strain YHYS, *Aspergillus tamarii* strain SRRC 1088, *Aspergillus niger* strain MCAS2, *Penicillium* sps. The isolated strains were tested for dye decolourization of Anthroquinone (textile) dyes. Anthroquinone (textile) dyes, (Reactive blue-19, Brilliant blue-R, Acid blue-25) are used at 100mg/L concentration (Devi and Kaushik, 2005). Sabouraud's dextrose broth was added separately with each of the Anthroquinone dyes and subsequently inoculated the fungal spore suspension containing 5.0×10^6 cfu/ml (colony forming unit) (K.R. Aneja, 2003). The inoculated flasks were incubated at $35\pm 1^\circ\text{C}$ for 8 days. After incubation the samples were centrifuged at 10000 rpm for 10 min decolourization efficiency was

assessed by measuring absorbance of the supernatant with the help of spectrophotometer at wave length maxima (λ_m) of respective dye (Remazol Brilliant Blue R(592nm), Brilliant Blue-R(470nm), Acid Blue-25(630nm)). Control flasks were maintained, experiments were performed (initial & final values) in triplicates results were expressed as the mean $\pm$ S.E (standard error) by IBM SPSS software 17.0 and the percentage of decolourization was calculated by using following formula:

$$\% \text{ of Decolourization} = \frac{\text{Initial OD} - \text{Final OD}}{\text{Initial OD}} \times 100.$$

3. Optimization of dye decolourization by environmental factors

3.1 Effect of temperature: Optimization studies were conducted by using different temperatures ranging from 25°C, 37°C, & 45°C with the known concentration of dyes (RB-19, BBR, AB-25) incorporated into the Sabouraud's Dextrose broth for decolourization. All the flasks were incubated under shaking conditions at 120 rpm for eight days and the results recorded in fig-3.a, 3.b, and 3.c.

3.2 Effect of pH: Decolourization of Anthraquinone dyes (RB-19, BBR, AB-25) with all six fungal species were optimized for pH ranging from 5.5, 7.0, 8.5 and controls were maintained. All the flasks were incubated under shaking conditions at 120 rpm for eight days (Fig-4.a, 4.b, and 4.c.)

4. Optimization of dye decolourization by Nutrients

4.1 Effect of Carbon sources: To enhance decolorization activity of RB-19, BBR, AB-25, culture medium supplement with different carbon sources such as Dextrose, Sucrose and Starch by isolated culture. All the flasks were kept for incubation at 37°C temperature under shaking conditions at 120 rpm for eight days (Fig-5.a, 5.b, 5.c).

4.2 Effect of Nitrogen sources: Optimization of decolorization activity by different Nitrogen sources such as Peptone, Yeast extract and Beef extract with isolated species and without inoculum was served as control. All the flasks were incubated at 37°C under shaking conditions at 120 rpm for eight days (Fig-6.a, 6.b, 6.c.)

5. Dye concentration

Optimization studies was carried out by different dye concentrations like 50mg, 100mg and 200mg with inoculum and without inoculum was served as control & incubated at 37°C for eight days and values were recorded.(Fig- Fig-7.a, 7.b, 7.c)

6. Shaking and Static conditions

The culture flasks were incubated at shaking and static condition for 100 ppm dye (RB-19, BBR, AB-25) with respective inoculum and control was maintained & incubated at 37° c under shaking and static conditions for eight days and decolorization potential was recorded.(Fig-8.a, 8.b, 8.c)

7. Results and Discussion

The fungal cultures isolated and identified from dye amended samples was observed under microscope and also they were sent to National Centre for Cell Sciences, Pune for 18S rRNA gene sequencing as per the reports obtained from NCCS the fungal species confirmed as *Aspergillus terreus*, *Aspergillus flavus* strain AJ, *Aspergillus fumigatus* strain YHYS, *Aspergillus niger* strain HP2A, *Aspergillus tamaraii* strain SRRC 1088, *Aspergillus fumigatus* strain SK1, *Aspergillus niger* strain MCAS2, *Aspergillus terreus* strain ATCC 1012, *Aspergillus niger* strain MCAS2, *Pencillium* sps.(Table-1).With the help of obtained sequences, they may be submitted to phylip data bases for phylogenetic tree. The derived phylogenetic tree was analysed for rational relationship among the species isolated with HMM values for their genetic distances.

Table:-1 the Sequenced fungal isolates and their accession numbers for each isolate is listed below.

S.no	Description	Accession number
1	<i>Aspergillus terreus</i>	<u>HQ219673.1</u>
2	<i>Aspergillus flavus</i> strain AJ	<u>HQ324118.1</u>
3	<i>Aspergillus fumigatus</i> strain YHYS	<u>KC986235.1</u>
4	<i>Aspergillus niger</i> strain HP2A	<u>KJ677240.1</u>
5	<i>Aspergillus tamaraii</i> strain SRRC 1088	<u>AY373870.1</u>
6	<i>Aspergillus fumigatus</i> strain SK1	<u>KM207771.1</u>
7	<i>Aspergillus niger</i> strain MCAS2	<u>KM103363.1</u>
8	<i>Aspergillus terreus</i> strain ATCC 1012	<u>AY373871.1</u>
9	<i>Aspergillus niger</i> strain MCAS2	<u>KM103363.1</u>
10	<i>Pencillium</i> sps	<u>MCC 1031</u>

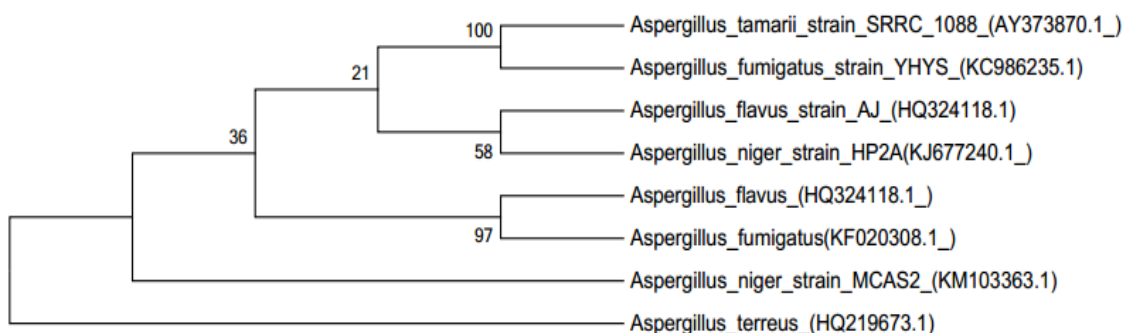


Fig:-1. Phylogenetic analysis was showing the phylogenetic relationships and closest related strains of the fungal isolates, based on 18S rRNA gene sequences, Bootstrap values were derived.

The isolated & identified fungal species *Aspergillus terreus*, *Aspergillus flavus* strain AJ, *Aspergillus fumigatus* strain YHYS, *Aspergillus tamaraii* strain SRRC 1088, *Aspergillus niger* strain MCAS2, *Pencillium* sps are very abundant & frequently available

industrial effluents and amended soils. The occurrence of fungi in the dye amended soils depends on the availability of nutrient, oxygen and biological, chemical and physical features of the contaminant.

The decolourization efficiency of these strains was monitored by measuring the optical density after 8 days of incubation. It was noticed that there should be decrease in the optical density by increasing the incubation time (Table-2). Among the six species *Aspergillus fumigatus* strain YHYS, *Pencillium* sps, *Aspergillus flavus* strain AJ, *Aspergillus niger* strain MCAS2, showed effective decolorization followed by *Aspergillus tamarii* strain SRRC 1088 and *Aspergillus terreus*. The fungal isolates *Aspergillus fumigatus* strain YHYS showed maximum decolorization 86.3% (AB-25), *Pencillium* sps, showed 84.9% (AB-25), *Aspergillus flavus* strain AJ showed maximum decolorization 83.0% (AB-25), *Aspergillus niger* strain MCAS2 showed decolorization potential 74.5% (RBBR), *Aspergillus tamarii* strain SRRC 1088 showed maximum decolorization 78.3%(RBBR) and *Aspergillus terreus* showed decolorization potential 53.1%(RBBR)(fig:-2).

There is a limited literature is available on *A. terreus*, *A. Fumigatus* about their applicability for decolourization of textile dyes. In the present study it was proved that most of the above mentioned fungi were showed the potential dye decolorization (*A. flavus*, *A. niger*). There are several advantages by using the fungal species for bioremediation, *Aspergillus flavus*(83.0%),*A. niger* (73.0%),*A.fumigatus* (86.3%)and *pencillium* sps(84.9%) proved to be efficient decolourizers of Acid blue-25 where as *Aspergillus tamarii* strain SRRC 1088 and *Aspergillus terreus* (50.8%) showed less potential of dye decolorization within 8 days of incubation.

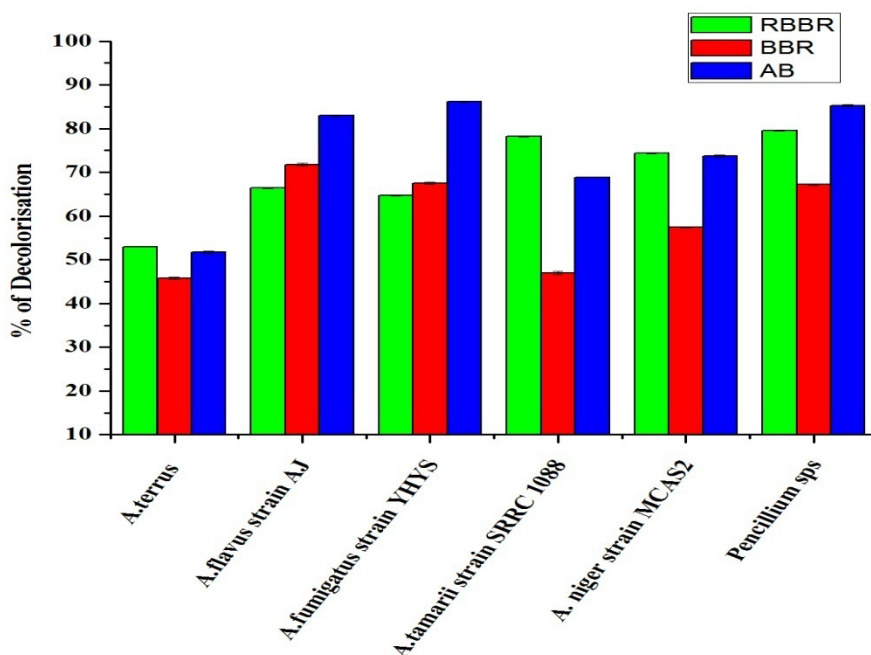


Fig- 2. Figure showing the % of dye decolorization by fungal species with Mean, standard deviation (S.D) and standard error (IBM SPSS 17.0) Statistics and graph was drawn by using origin-8.0 software. Values are indicated in the table mean $\pm$ S.E (Standard Error).

Temperature is an impotent physical parameter which effectively contributes its tendency on the influence of metabolism (growth & enzyme) of the fungal species used for dye decolorization activities. In the present study, we used different temperature ranges from 25⁰c, 37⁰c, 45⁰c and their optimization were calculated. Three different temperatures were used for conducting the efficiency of decolorization activity using RB-19, BBR, AB-25 dyes by fungal strains. From the results it was noticed that the maximum decolorization 86.3% was observed at 37⁰c in AB-25 by *A. fumigatus strain YHYS* while increasing the temperature decreases the decolorization activity. (Fig-3.a,3.b, 3.c).

Reactive blue-19

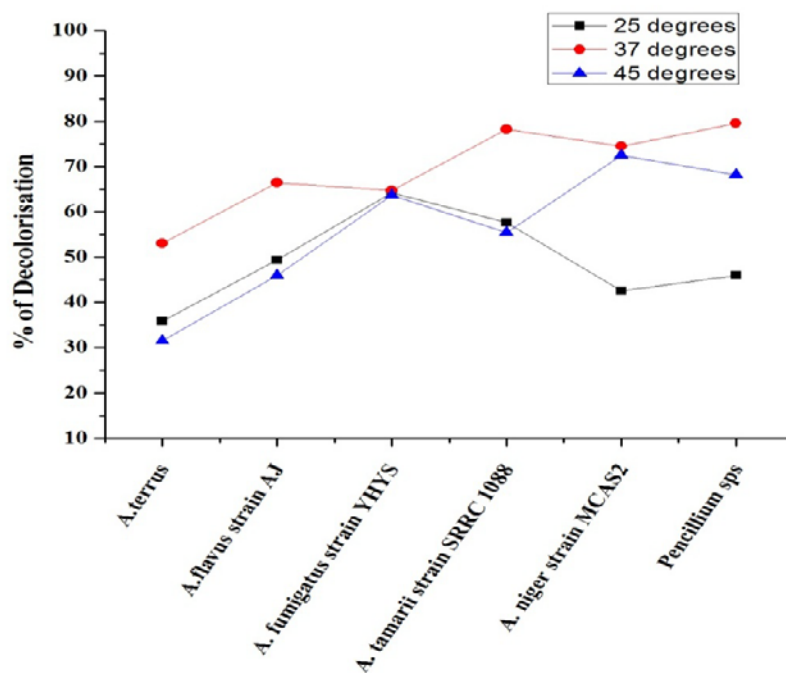


Fig-3.a. Effect of temperature on Reactive blue-19 dye decolorization.

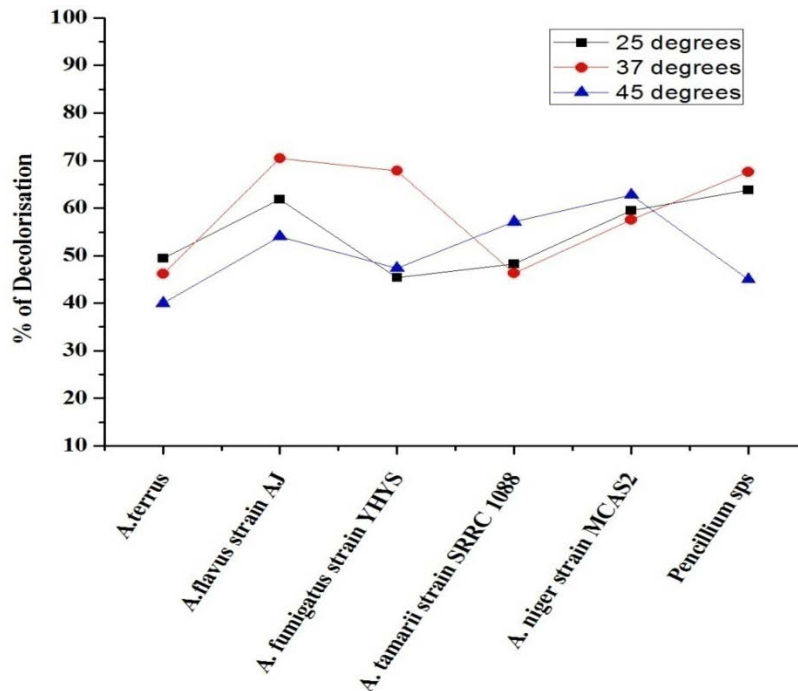


Fig-3.b. Effect of temperature on Brilliant blue-R dye decolorization.

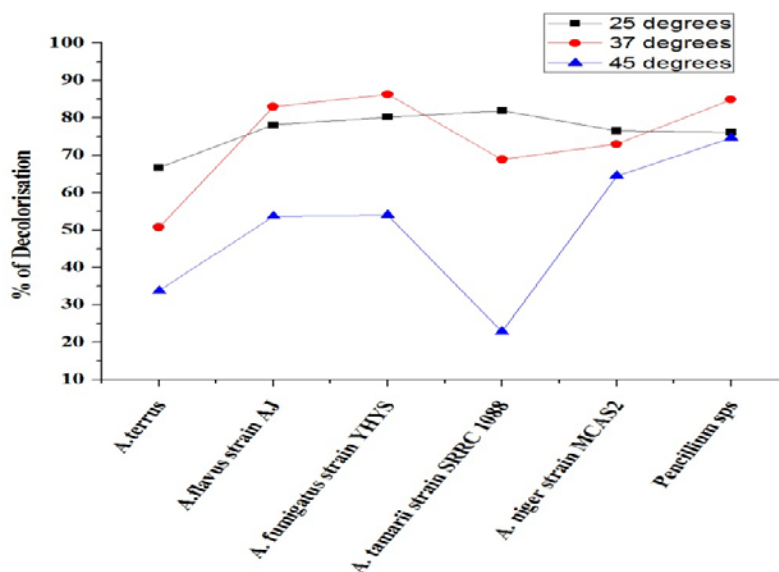


Fig-3.c.Effect of temperature on Acid blue-25 dye decolorization.

P^H is an important environmental factor for the growth and metabolic activity of the fungal organisms, it is differ from one species to other species. To investigate the effect of p^H on Anthraquinone dye (RBBR.BBR, AB-25) decolorization activity by fungal strains. Here we are using different pH ranges from 5.5, 7.0 & 8.5 and their optimization was calculated. Maximum decolourization activity 86.3% was observed by *A. fumigatus* strain YHYS at p^H -5.5 and minimum decolourization was observed 28.5% by *A.terrus* at pH-8.5 (fig-4.a, 4.b, 4.c).

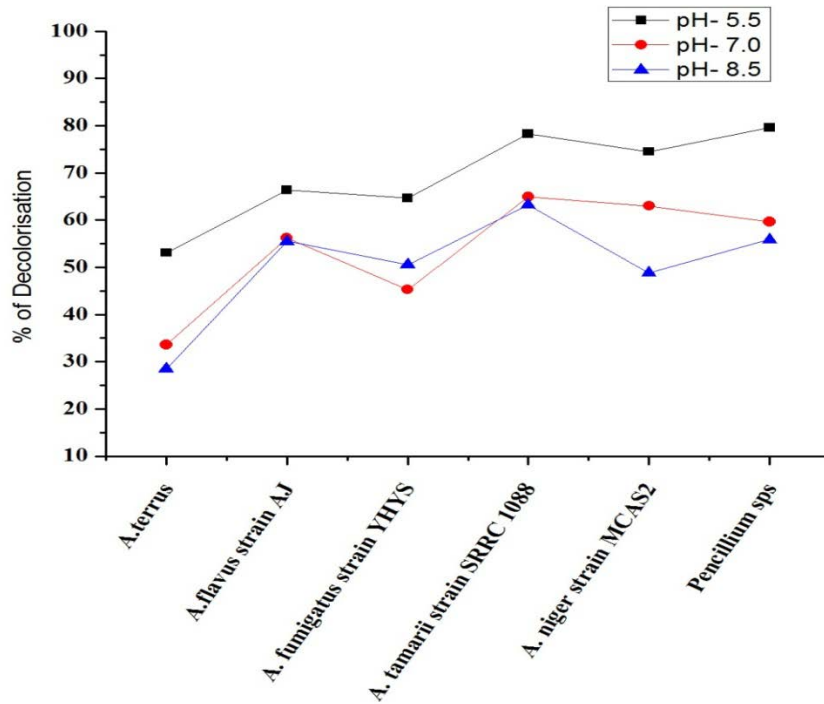


Fig-4.a.Effect of pH on Reactive blue-19 dye decolorization.

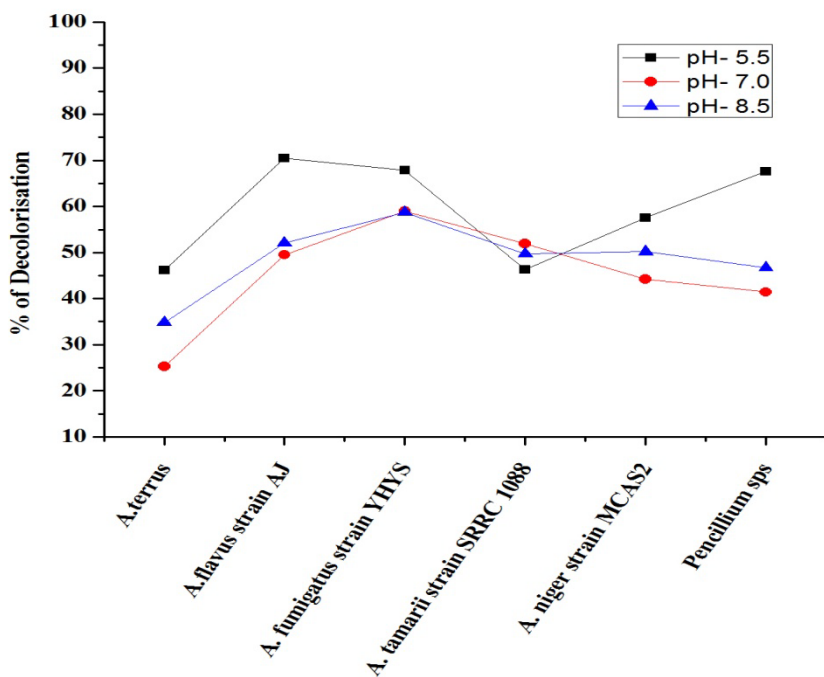


Fig-4.b. Effect of pH on Brilliant blue-R dye decolorization

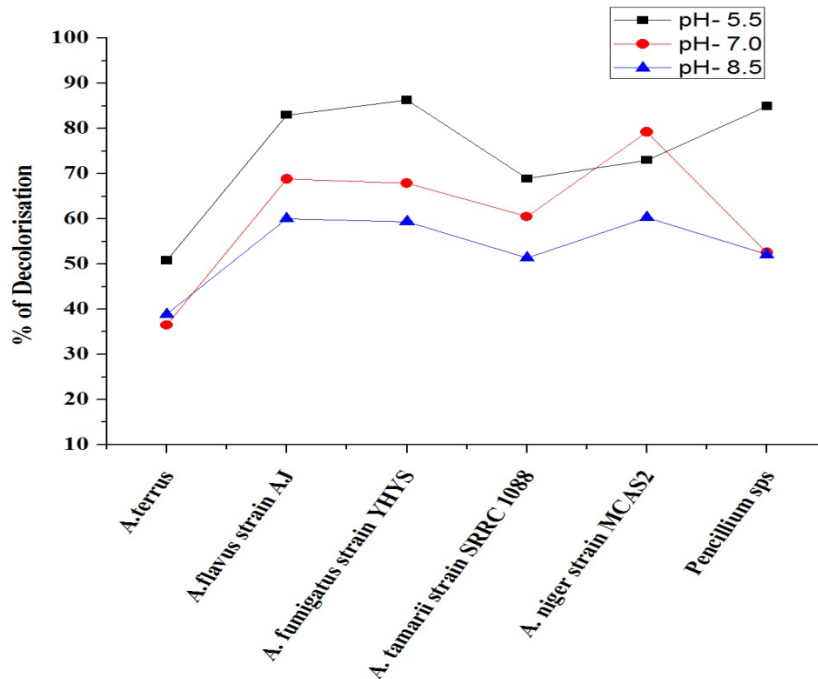


Fig-4.c.Effect of pH on Acid blue-25 dye decolorization.

Optimisation of dye decolourization efficiency was enhanced by supplementing with different carbon sources like Dextrose, sucrose and starch in Saubourds broth by fungal isolates. When dextrose and sucrose was supplemented in the broth showed the potential decolorization more than 80 % of AB-25 (100 mg/l) was observed by *A.fumigatus strain YHYS* and *Pencillium sps*, *A.flavus strain AJ* and minimum decolorization efficiency 38.5% (RBBR) was observed when starch used as carbon source by *A.terrus* within 8 days of incubation under shaking conditions (fig-5.a, 5.b,5.c).

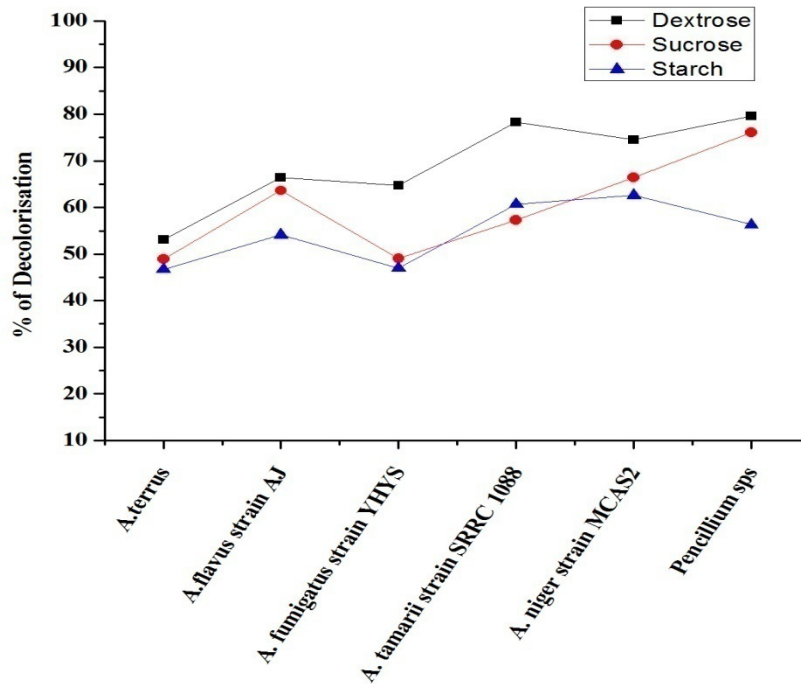


Fig-5.a. Effect of carbon sources on Reactive blue-19 dye decolorization.

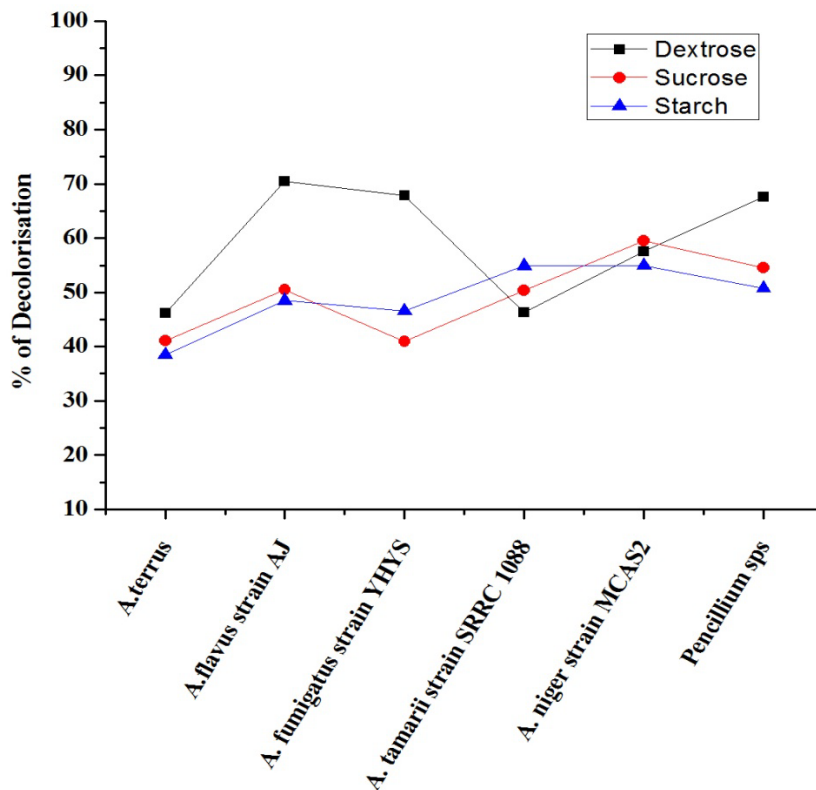


Fig-5.b. Effect of carbon sources on Brilliant blue-R dye decolorization

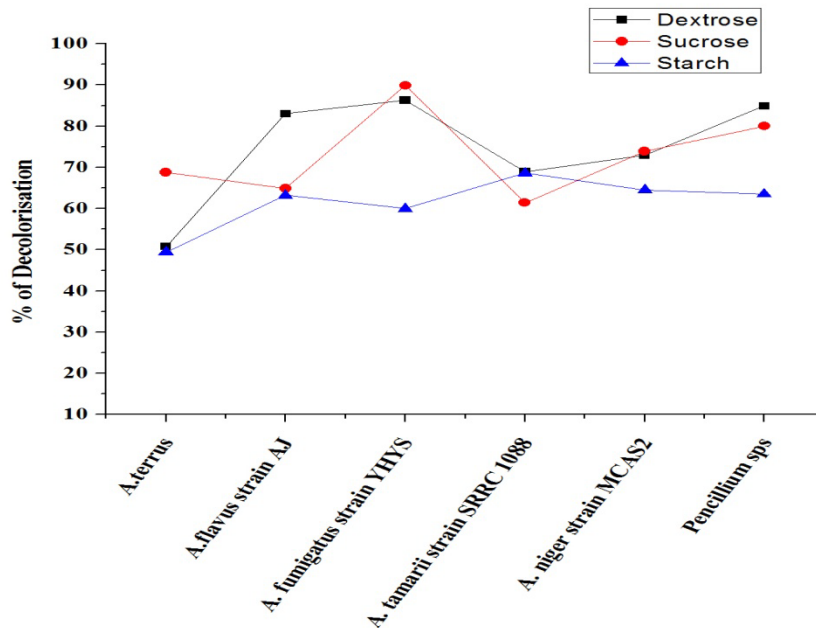


Fig-5.C. Effect of carbon sources on Acid blue-25 dye decolorization

Nitrogen is an important nutritional factor which effectively influences the metabolic activities of the organisms. Different nitrogen sources have been used for decolorisation activity such as Peptone, Beef extract and Yeast extract. The decolourization activity of Acid blue-25 with yeast extract showed 81% by *A. tamarit strain SRRC 1088*, with peptone showed 86.3%(AB-25), 84.9% (AB-25) and 83.0% (AB-25) by *A. fumigatus strain YHYS*, *Penicillium sps* and *A.flavus strain AJ* respectively. The minimum decolorisation was recorded with RB-19(16.4%) by *A.terrus*(fig-6.a, 6.b, 6.c).

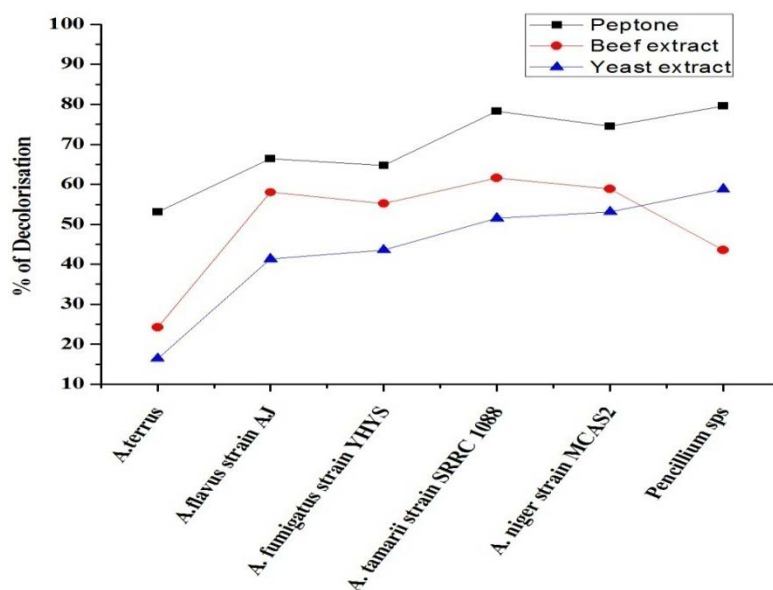


Fig-6.a. Effect of nitrogen sources on Reactive blue-19 dye decolorization.

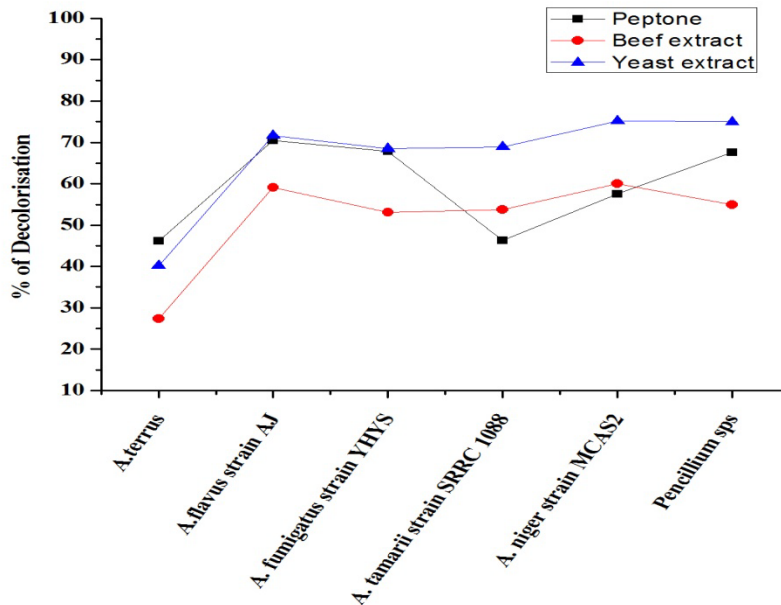


Fig-6.b. Effect of nitrogen sources on Brilliant blue-R dye decolorization.

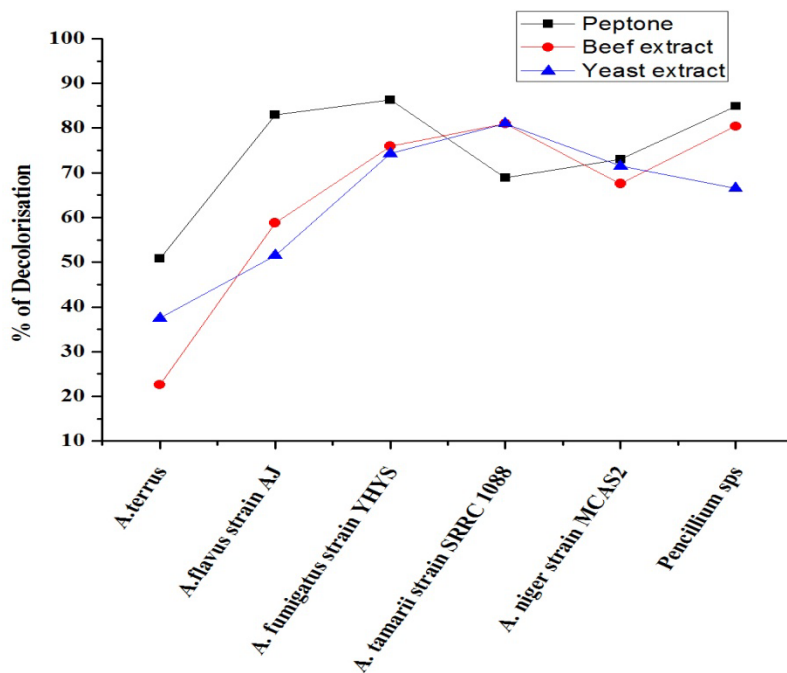


Fig-6.c. Effect of nitrogen sources on Acid blue-25 dye decolorization

Optimization of fungal strains decolourization activity was studied by using different concentrations of textile dyes like Remazol brilliant blue-R, Brilliant blue-R and Acid blue 25 varying concentration from 50 mg to 200 mg. The decolourization rate was

high at 50ppm concentration and then decreased by increasing dye concentrations (100ppm, 200ppm) with inoculums. If further dye concentration was increased the decolorization activity decreased (fig-7.a, 7.b, 7.c).

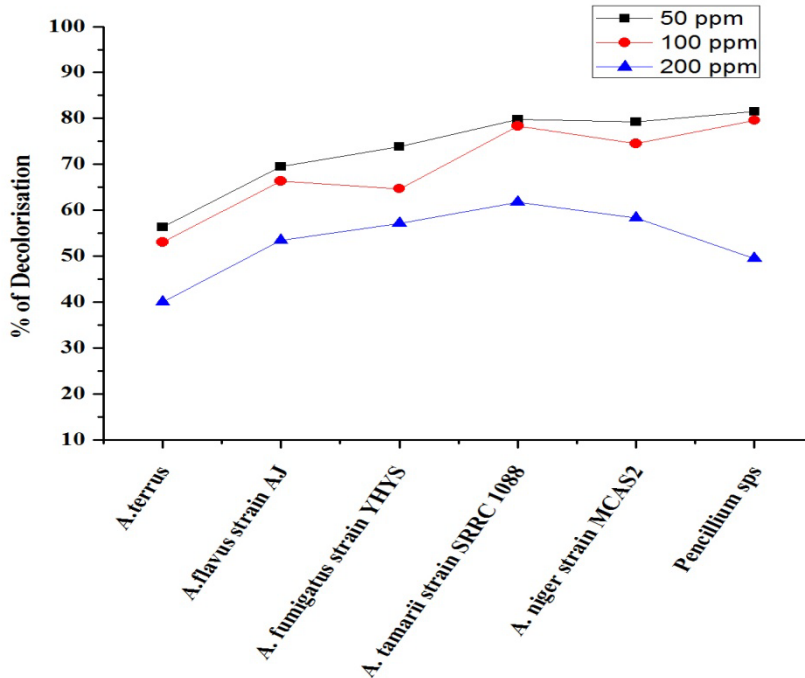


Fig-7.a Effect of dye concentration on Reactive blue-R dye decolorization rate.

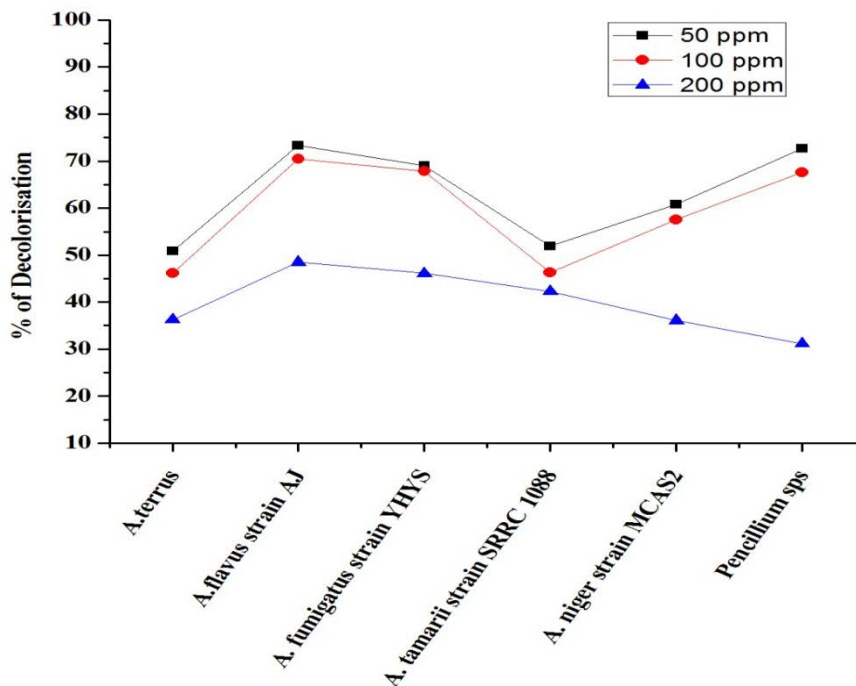


Fig-7.b Effect of dye concentration on Brilliant blue-R dye decolorization rate.

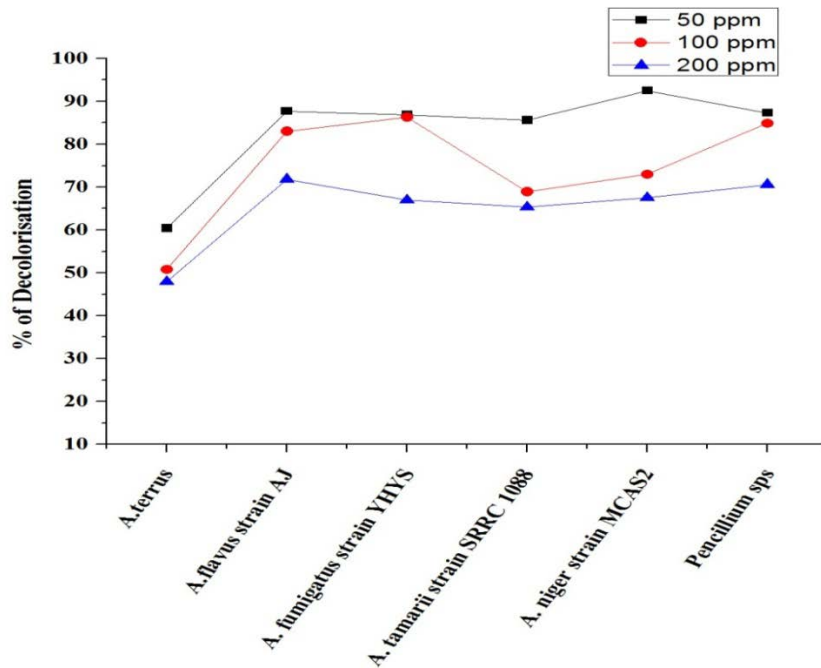


Fig-7.c Effect of dye concentration on Acid blue-25 dye decolorization rate.

Decolourization activity of fungal organisms was observed under shaking and static condition. All the fungal species noticed the maximum decolorization activity in shaking condition when compared to static condition. The potential decolorisation was observed 86.3% at shaking and 55.3% at static condition by *A. fumigatus* strain YHYS. The minimum decolorization was observed 16.9% (RBBR) under static condition by *A.terrus*(Fig-8.a,8.b,8.c).

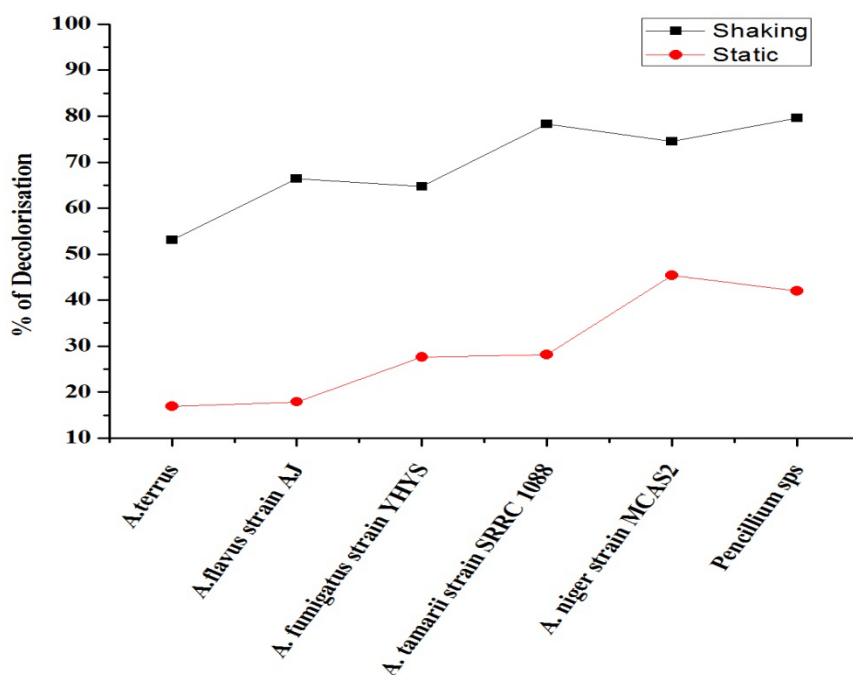


Fig-8.a. Effect of shaking & static condition for Reactive blue-19 dye decolorization.

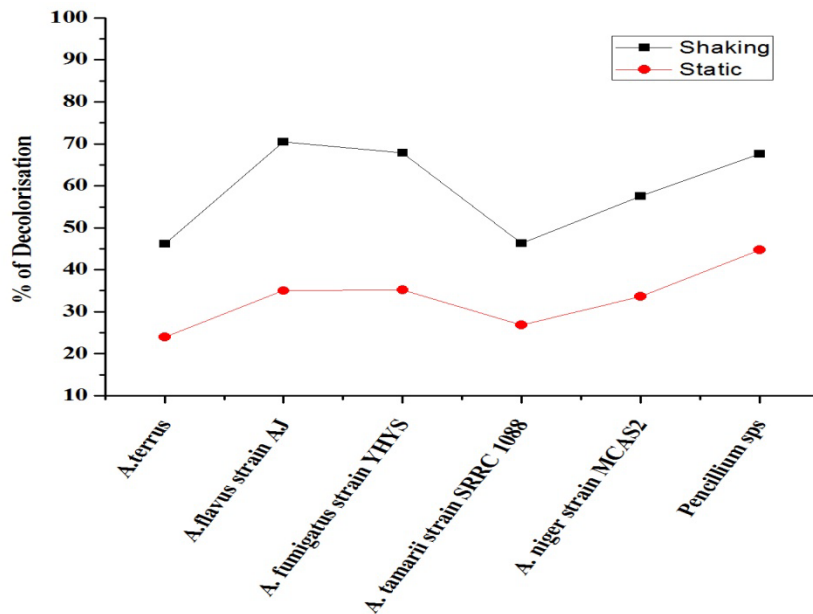


Fig-8.b. Effect of shaking & static condition for Brilliant blue-R dye decolorization.

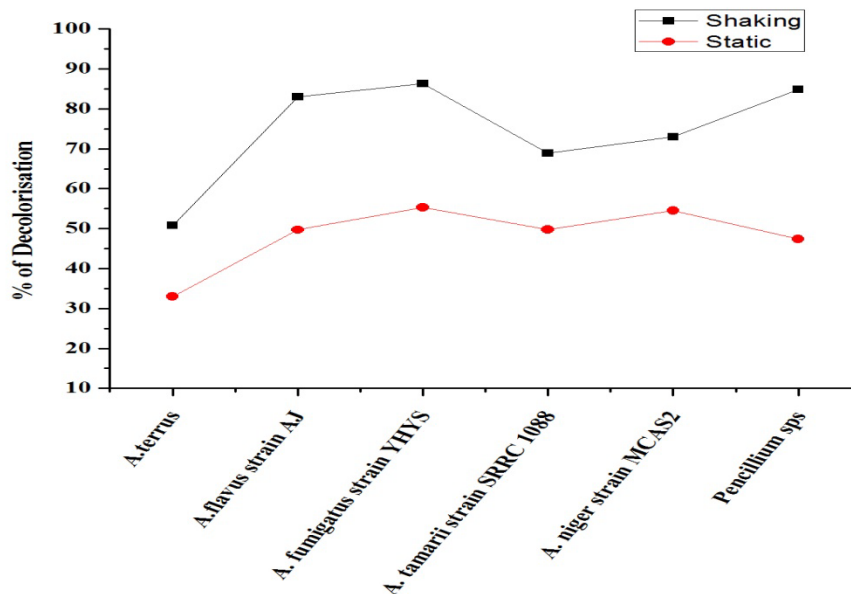


Fig-8.c. Effect of shaking & static condition for Acid blue-25 dye decolorization.

The mechanism of decolourization may be observed by method biosorption, which is dependent on functional groups in the dye molecule and amount of fungal biomass which may also be playing role in the biosorption of dye (Fu and Viraraghavan, 2002). Under aerobic condition oxido reductases usually cleave Anthroquinone dyes in to the corresponding amines, many of which are mutagenic and/or carcinogenic (O' Neill et

al., 2000; Chivukula and Renganathan, 1995; Chung and Stevens, 1993). Excess production of protein in response of dyes and growth in C-limited medium indicated that the fungus utilized the dyes as the sole source of carbon and produced enzymes to degrade the dyes.

Similar by present study mainly focused on decolourization of Anthroquinone textile dye(RBBR,BBR,AB-25) by different fungal species, Kousar and Charya (2002) also reported for fungi such as *Aspergillus.flavus*, *A.niger*, isolated from textile and dye contaminated soils were able to decolorize the dye effluent (Nicola et al., 1998). In the biotechnological approach the colour removal technique was suggested for microbiological treatment is an ideal method for colour removal where less sludge is produced in comparison to lower daily running cost on anchored in comparison to chemical treatment.

6. Conclusions

In the present study the six isolated & identified fungal species could be subjected for decolorization activity by using shaking condition. Further their efficacy of % of decolorization was enhanced through several physical, environmental, nutritional factors. Optimization was also recorded with maximum potentiality that ensures the dye tolerant capacity of an organism. The ability of the strain to decolorization and degrade the Anthroquinone dyes at high concentration has advantage to the treatment of textile effluents.

7. Acknowledgments

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8. Reference

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